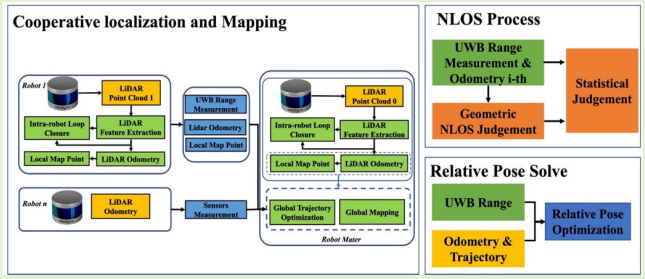
Range-Aided LiDAR Cooperative Localization and Mapping

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***Abstract*—The localization and mapping of the robotics team in unknown environments, especially GNSS denied environments, is a prerequisite for the perception and exploration of unknown environments by the robotics team. In this article, we propose an algorithm based on sliding window graph optimization, which incorporates ultrawideband (UWB) range information and LiDAR odometry information for real-time localization and mapping. The cooperative localization mapping system mentioned in this article incorporates a non-line-of-sight (NLOS) error discrimination method based on triangulation and statistical fusion. Based on Fisher information matrix, we investigate the number and arrangement of UWB ranging sensors to compensate for the drift of the LiDAR odometry. Finally, experiments on real scenes confirm the precision of our proposed co-localization system and the collaborative mapping of real scenes. The effectiveness of the fusion of UWB ranging information with LiDAR odometry information and the ability of ranging information to correct for LiDAR odometry drift are demonstrated.**



***Index Terms*— Cooperative localization mapping, information fusion, range-aided, ultrawideband (UWB).**

I. Introduction

IN SIMULTANEOUS localization and mapping (SLAM), LiDAR is widely used for its high-accuracy measurement. At the same time, detection and mapping of team robots in unknown environments has been developed as an application of simultaneous localization and mapping. Collaborative pose estimation between different robot platforms can improve the accuracy of odometry estimation between different robots [1]. In addition, the detection of environments without a priori information by different multirobots has a stronger detection capability than that of a single robot, which is mainly reflected in the wider detection range the same time, more detection data obtained, and most importantly, the collaborative detection enables the collaborative use of detection information between different robots [2]. One of the key problems to be solved for cooperative pose estimation is the information of cooperative multirobot pose measurements. To achieve this problem, requiring not only the ego-motion poses of each robot but also the information of relative pose about the multirobot is required to know. Stable relative positional estimates are the basic information needed for coordinate consistent collaborative real-time mapping.

SLAM based on camera and LiDAR can perform effective self-localization and real-time mapping in GNSS-denied environment, especially the loop detection based on co-view area effectively reduces the drift of robot trajectory estimation. The loop detection based on co-view area can be applied to cooperative localization and mapping teams to achieve relative position estimation among different robots in the robot team. Relative positional estimation based on loop detection has limitations in both the use of scenarios and the effective time: first, even at locations where co-viewing regions should not occur, they may be misclassified as co-viewing regions due to the similarity of the scenarios; second, relative poses are generated between different robots only at locations where co-viewing exists, so the number of such estimates generated

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tends to be small, and therefore, the relative poses generated become less reliable and effective as time progresses.

The use of ultrawideband (UWB) ranging to generate solutions for the poses in robotic teams is an effective method for solving using co-visual regions. This approach has benefited from the emergence and maturation of wireless sensing localization and navigation theory. In a GNSS-denied environment, UWB ranging using a fixed base station allows the robot to obtain a global position and thus achieve globally consistent localization and thus coordinate consistent environmental awareness. In addition, anchors-free UWB ranging enables relative position constraint between different robots to obtain relative pose measurements between different members and global consistent pose estimation using ego-pose estimation of different robots and relative pose measurements between them. Obviously, anchor-free UWB ranging is more flexible in the arrangement of robot teams.

In this article, we propose a method for unifying relative measurement information between different robots and their own local positional estimation information into a global coordinate system through distance measurement fusion of LiDAR odometry and anchor-free UWB. The main contributions of this letter are summarized as follows.

1) Our approach utilizes a sliding window optimization technique for cooperative position estimation and consistent point cloud map construction, which effectively addresses non-line-of-sight (NLOS) issues.
2) Additionally, we construct a consistent point cloud map through global consistent pose-map construction without global co-vison.
3) Through simulations and practical experiments, we demonstrate the effectiveness of UWB alignment on the robot's positional estimation and its ability to suppress errors in the direction of estimated drift with insufficient observational information.

II. Related Work

A. Relative Localization

In this article, we discuss and focus on the relative poses for public observation area matching and the relative poses solution algorithm for UWB ranging constraints. The observation of objects in the common field of view of different robots by joint adjacent moments or the same moments of common area observation often requires camera information or LiDAR information [3], [4], [5], [6], [7]. They solve the relative poses by extracting geometric information, such as feature points or feature surfaces in a common area. Jung et al. [7] demonstrated the observability of this method for solving relative poses. However, this method must require an overlap of the observation areas of different robots, which requires that the observation field of view of the camera or other observation sensors is sufficient. In some other works [3], [8], [9], [10], [11], relative pose solution is achieved by means of target detection. Inevitably, the co-view area-based approach suffers from field-of-view limitations, while the target detection-based approach suffers from false recognition and is computationally more expensive.

Another solution is based on wireless range that does not rely on environmental information, such as UWB with low power and high range accuracy. As the accuracy of UWB ranging improves, the UWB ranging information is fused into the relative positioning or relative state estimation of inter-robot. The validity of the information used for fused UWB ranging has a huge impact on the accuracy of the relative positioning or relative state estimation of the inter-robot. Provided that the measurement uncertainty due to clock synchronization is known, we classify the relative positional estimation methods based on UWB ranging into single tag ranging [12], [13], [14], [15] and multitag ranging [16], [17], [18], [19], [20] according to the number of ranging tags. Relative position measurement based on UWB ranging has the advantages of high measurement frequency, no observed field of view limitation, and small amount of transmitted data compared to the common field of view matching-based approach. Single-tag-based relative pose measurement method can cause the relative pose to be unobservable when the two robot nodes have special forms of motion, such as parallel motion. Jones et al. [18] adapted that the relative positional estimation to the presence of NLOS by constant velocity model of assumptions and multiple UWB ranging model observations through extended Kalman filter (EKF) makes the relative positional estimation adaptable to the presence of NLOS. Shalaby et al. [19] achieved 3-D pose and position observable by fusing IMU and multiple UWB tag ranging and completed the relative positional estimation using EKF method. Fishbergp and How [20], by fusing odometry and multiple UWB ranging tags, solved the relative positional estimation in the 2-D case based on a graph optimization model.

B. Collaborative Localization and Map Fusion

During the exploration of the GNSS-denied environment by the robotics team, obtaining its own position and environment reconstruction are two important tasks. The main current approach is carried out by constructing pose graph optimization (PGO) problems. We distinguish between vision-based/point cloud co-viewing type and distance measurement-based type according to the type of co-perception information. The first is a system of collaborative localization and mapping through cameras or LiDAR, which build and patch together the environmental structure of the environment, in which the robot team is located through co-viewing range [4], [21], [22], [23], [24]. Karrer et al. [21] introduced an inertial measurement unit (IMU), which solves the scale problem not addressed. Schmuck et al. [22] improved the communication module, allowing for a larger number of nodes to collaborate, but the method was not tested in the field in real time. They all require a powerful central processor unit. Karrer et al. [21] used trajectory alignment in such a way that individual robots observe overlapping regions at their initial positions, which allows trajectory alignment to be resolved at the initial moment of system operation. Zhang et al. [24] used binocular vision and PGO fusion to fuse the trajectories of multiple robots and fuse the map information of each robot node based on trajectory alignment and then completed collaborative mapping.

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Bernreiter et al. [25] addressed a multirobot framework for LiDAR based on the PGO framework proposed by Deutsch et al. [26] for multirobot cooperative localization. The high-precision 3-D measurement of LiDAR can reach the centimeter level, so LiDAR is superior to vision measurement for 3-D information acquisition. Dubé et al. [27] constructed a centralized communication network that can transmit all robot point clouds, and the existence of this network allows the 3-D LiDAR build information and the positional information of each robot node in this local area network to achieve a collaborative and consistent build on the central processor. Xie et al. [28] changed the communication method to peer-to-peer communication. Huang et al. [29] used scan context (SC) as a distributed data transfer between different robots to reduce the amount of information transfer, while Zhong et al. [30] used loop detection on a single robot node, which is based on position solution, which reduces the failure of loop detection to some extent. This reduces the error of loop detection to some extent. However, all the above methods are validated by the existence of overlap in the exploration area. Considering the effective matching characteristics of LiDAR, the above methods require a wide overlap region to satisfy the reliable relative pose constraint.

However, both LiDAR and vision-based collaborative mapping require a common observation area to align the coordinate systems of the members of the robot team. Moreover, their information transmission bandwidth is a great test for the network capacity of the robot team. UWB ranging-based co-localization systems have the advantages of less data transmission bandwidth, higher measurement frequency, and noncommon-view constraints. Papadimitriou et al. [31] achieved the co-localization of ground and aerial robots by fusing multiple UWB ranging units and a 9-axis IMU. This work fuses the two types of information through ESKF, which has the advantage of high update frequency compared to EKF. Liu et al. [32] achieves distributed cooperative localization of two vehicles by fusing odometer and UWB and makes NLOS determination by observation and prediction, and they achieve distributed cooperative trajectory estimation by distributed PGO. The above work is based on dynamic UWB anchor points. When a UWB base station is present, the UWB base station provides a priori information for the ranging information, i.e., there is no need to estimate the location of the UWB base station. Song et al. [33] fuses the ranging information from the base station with 2-D LiDAR data by means of an EKF. Perez-Grau et al. [34] fused RGBD cameras with UWB ranging information from a fixed base station, and the system was adapted to a multi-UAV robotics team, in which the RGBD cameras enable depth image-based global repositioning at low UAV flight altitudes. Cao and Beltrame [35] tightly coupled UWB ranging information from a single base station with a VINS system, which can theoretically achieve coordinate alignment of a multirobot system. Zhou et al. [36] aligned the UWB ranging information of multiple base stations with the local coordinate system of a single robot in a way that enables coordinate consistent cooperative map building and trajectory estimation for multirobot groups in this framework.

Therefore, we consider that UWB-based ranging is easier to obtain lightweight, high-frequency, effective relative pose measurement information compared to co-visual-based constraints. In addition, dynamic UWB anchor is easier to obtain compared to static UWB base stations, which require precise placement, which is inflexible on the one hand, and their positions are prone to deviation on the other hand. Therefore, we use dynamic UWB anchor for obtaining relative measurement information. The LiDAR odometry shows excellent capabilities in both mapping and localization, so we use UWB fused with LiDAR odometry. We use a stepwise optimization approach to first construct relative pose optimization and solve for the poses of each member of the robot team based on the relative pose optimization.

III. Methods and Numerical Results

Fig. 1 illustrates the collaborative localization and mapping of a robot swarm using LiDAR odometry and UWB ranging information. Each robot node transmits the local map and the estimated pose in its local coordinate to the central node in real time. The central node collects UWB ranging information in addition to the local map and local pose estimation information of each robot node. The global map construction and global pose estimation are done at the central node. In the rest of Section III, we present the details of our co-localization and map building system separately. Section III-A describes the variable representation that is needed to describe the coordinate system and its representation in this article. Section III-B describes the co-localization process. Section III-C describes the initialization process of relative pose estimation and NLOS processing during co-localization and map building. Section III-D describes the co-map building process. Section III-E describes the effect of the UWB tags placement problem on the relative pose estimation accuracy.

A. Problem Formulation

We consider the robot team has n robots, so the robot team can be represented as $\{R_i\}, i = \{0, 1, \ldots, n-1\}$, where R_0 is used to represent the robot master. The local coordinate system of each robot node is represented as $\{L_i\}, i = \{0, 1, \ldots, n-1\}$. We consider the pose at the initial moment of R_0 running, i.e., $t = 0$ moment as the global coordinate system $\{W\}$. In addition, we assume that the coordinate system of each robot node body is $\{B_i\}, i = \{0, 1, \ldots, n-1\}$. Therefore, we denote the pose and position of R_i under the coordinate F at moment t as ${}^{F}R_i^t \in \mathrm{SO}(3)$, ${}^{F}p_i^t \in R^3$. At moment t, the spatial transformation relation for the transformation of coordinate L_i to coordinate L_j can be expressed as ${}^{L_i}R_{L_j}^t \in \mathrm{SO}(3)$, ${}^{L_i}P_{L_j}^t \in R^3$. For R_i's mth UWB tag and R_j's nth UWB tag at moment t and the position under the coordinate F can be denoted as ${}^{F}P_{i,m}^t$ and ${}^{F}P_{j,n}^t$, respectively.

For the problem mentioned in this article, we assume that the variables to be estimated are expressed as the state variables of each robot in the global coordinate system $\{W\}$ as

$$\begin{Bmatrix} {}^{W}X_0 & \cdots & {}^{W}X_{n-1} \end{Bmatrix}, \quad {}^{W}X_i = \begin{Bmatrix} {}^{W}X_i^1 & \cdots & {}^{W}X_i^t \end{Bmatrix}$$
$$ {}^{W}X_i^j = \begin{Bmatrix} {}^{W}R_i^j & {}^{W}P_i^j \end{Bmatrix} \tag{1}$$

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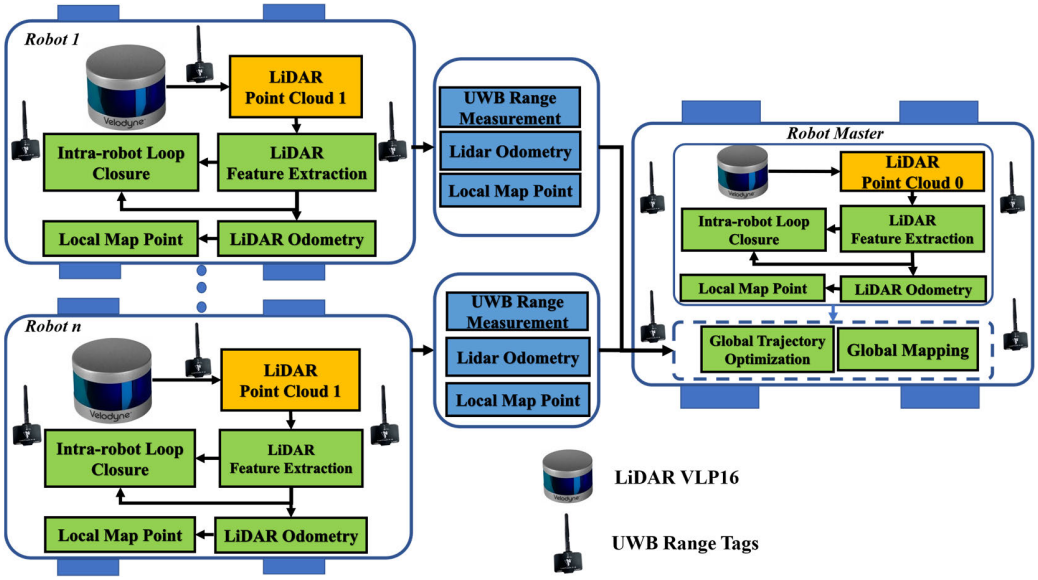


Fig. 1. System architecture. Each robot equips with a LiDAR, where the robot master has four UWB range antennas and the other robots have three range tags. Each robot sends its own local trajectory and local map information to the robot master, which is responsible for collaborative trajectory estimation and collaborative mapping. In our system, four UWB tags on the robot master is the UWB anchors, and the robot nodes have three UWB Tags. Each robot has one VLP16 LiDAR.

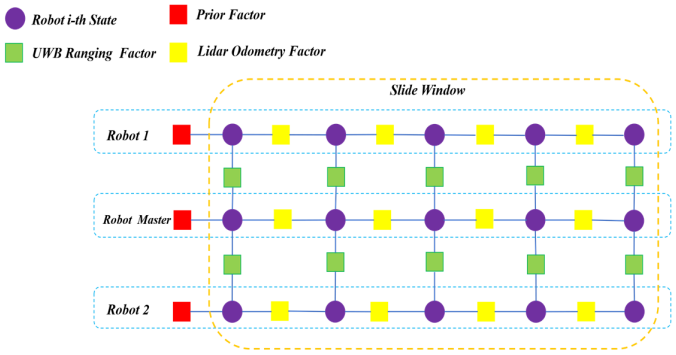


Fig. 2. Cooperative localization factor graph. The factor graph is optimized using a sliding window optimization, so the constraints that make up the factor graph consist mainly of LIDAR odometry constraints and UWB ranging constraints and a priori factors.

where we denote the estimated value of some variable x as $\widehat{x}$, i.e., $\widehat{{}^{L_i}P^t_{L_j}}$. We define here, in general terms, the observation during the running of the system as $Z = \{ Z^1 \cdots Z^t \}$. We formulate the cooperative trajectory estimation as the maximum a posteriori (MAP) problem: $\widehat{X} = \text{argmax}_X P(X|Z, {}^W X^1_0)$.

B. Cooperative Localization

Fig. 2 illustrates the factor diagram of the cooperative pose estimation. We use a sliding window filter to fuse the UWB range information and LiDAR odometry data to batch estimate the trajectory estimation information over a period. The measurements from the UWB and the LiDAR odometry at moment $1{:}t$ are $\{ Z^1 \cdots Z^t \}$, and the measurements at moment i are $\{Z^i_{\text{uwb}}, Z^i_{\text{Odo}}\}$. Then, the MAP problem: $\widehat{X} = \text{argmax}_X P(X|Z, {}^W X^1_0)$ converts to the sliding window filters of length $M = t - m + 1$ constructed at moments $1{:}m-1$ and $m{:}t$

$$\left\{ \overbrace{X^1 \quad \cdots \quad X^{m-1}}^{\text{prior}} \qquad \underbrace{X^m \quad \cdots \quad X^t}_{M} \right\}. \tag{2}$$

The probability density function (pdf) of the sliding window filter can be expressed as

$$P\left(X^{m:t} \middle| Z^{m:k}, X^1\right) = \eta P\left(X^{m:t} \middle| Z^{m:t}, X^m\right) \times P\left(X^m \middle| X^1, Z^{1:m-1}\right). \tag{3}$$

It is worth noting that the UWB measurements and the output of the LiDAR odometry are not perfectly aligned most moments. We would like to utilize not only the full observational information at the moments when the UWB measurements and LiDAR are perfectly aligned but also to utilize as many UWB measurements and LiDAR odometry data with similar measurement moments as possible. We use linear interpolation to obtain LiDAR odometry measurements that are closer to the UWB measurement moment. The specific description of the interpolation will be described specifically in Section III-B3.

1) UWB Range: The concrete position of the mth UWB Tag of the ith robot member at moment t in the global coordinate $\{W\}$ is expressed as ${}^W P^t_{i,m}$. We assume that the attitude and position of this robot in the global coordinate are ${}^w R^t_i$ and ${}^W P^t_i$, respectively. The coordinates of this UWB tag under the robot body coordinate system $\{B_i\}$ are ${}^{B_i} P_{i,m}$. Then, the coordinates of this UWB tag in the global coordinate system $\{W\}$ have the expression relation as shown in the following equation:

$${}^W P_{i,m} = {}^W R^t_i \times {}^{B_i} P_{i,m} + {}^W P^t_i. \tag{4}$$

Assuming range measurement noise follows an approximately normal distribution $n_r \sim N(0, \sigma_r)$. The distance measurement ${}_{\text{UWB}} Z^t_{im,jn}$ between the mth UWB tag of the ith robot and the nth UWB tag of the jth robot at moment t is modeled as

$$\begin{aligned} {}_{\text{UWB}} Z^t_{im,jn} &= \left\| {}^W P^t_{i,m} - {}^W P^t_{j,n} \right\| + n_r \\ &= \left\| {}^W R^t_i \times {}^{B_i} P_{i,m} + {}^W P^t_i - {}^W R^t_j \times {}^{B_j} P_{j,n} - {}^W P^t_j \right\| + n_r. \end{aligned} \tag{5}$$

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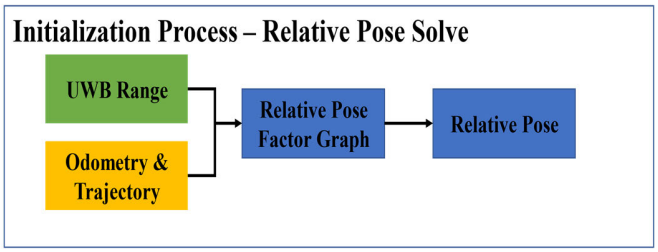


Fig. 3. Initialization flow diagram shows that the UWB range information and odometry are collected on robot master's central processor for the initial relative position solution.

It is to be noted that our system uses an aggregate system, and our UWB ranging only measures the distance between robot master's tags and the UWB tags of other robot nodes. In addition, we assume that our system is suitable for the TWR-TOA measurement mode of UWB. The reason for this approach is that the two-way communication UWB ranging mode makes the clock synchronization error less affected by the measured distance. Therefore, it is reasonable to believe that the ranging noise is more consistent with a certain specific Gaussian measurement noise.

2) LiDAR Odometry: Considering the better estimation performance of the LiDAR odometry, the odometry measurement output used in the system is the LiDAR odometry. What needs to be employed in the system is the relative poses of the LiDAR odometry at adjacent moments as the measurement of intra-robot. Assume that the attitude and position of robot i under its local coordinate frame L_i at moment t are represented as ${}^{L_i}R_i^t$ and ${}^{L_i}P_i^t$ and ${}^{L_i}R_i^{t-1}$ and ${}^{L_i}P_i^{t-1}$ at moment $t-1$. Then, the relative positional measurements ${}_{\text{odo}}Z_i^t$ at moment t are

$$ {}_{\text{odom}}Z_i^t = \begin{cases} {}^{L_i}R_i^{t-1^T}\,{}^{L_i}R_i^t \\ -{}^{L_i}R_i^{t-1^T} \times {}^{L_i}P_i^{t-1} + {}^{L_i}R_i^{t-1^T} \times {}^{L_i}P_i^t \end{cases} + n_{\text{odom}}. \quad (6) $$

We consider the operation of the LiDAR odometry as a first-order Markov process with state ephemerality at stable observation. Therefore, we assume that the noise n_{odom} of the odometry observation of the relative positional is a Gaussian noise, $n_{\text{odom}} \sim N(0, \sum_{\text{odom}})$.

3) Optimization-Based Relative Pose Estimation: The robot team network mentioned in this article is a centralized network, where the measurement data are transmitted to robot master via LAN. The UWB range information and the odometry data from each robot node as well as the local map information are transmitted to the central node. Fig. 3 illustrates the process of sliding window optimization for the observations transmitted to the central node. Equations (2) and (3) represent the variables used for estimation and the posterior probability distribution composed of these variables and observations, respectively. The term $P(X^m|X^1, Z^{1:m-1})$ in (3) yields the third constraint a priori constraint in this part. The observations added to the cooperative pose estimation are not all the observations. To ensure the validity of the observations, we select some of the key frames as the real observations to be added to the optimization.

There are two types of measurements that become keyframes: when the UWB measurement is LOS, the keyframes are obtained by interpolation; when the UWB measurement exists NLOS, the LiDAR odometry is obtained by solving for the point cloud constraints that need to be large adequate. In the LOS case, the UWB is acquired at moment t. For any robot i, the odometry acquisition moments close to t are $t_1 < t < t_2$, respectively, and we construct an interpolation based on the degree of distance from moment t

$$ {}^{L_i}R_i^{t_1} \Leftrightarrow {}^{L_i}q_i^{t_1}, \quad {}^{L_i}R_i^{t_2} \Leftrightarrow {}^{L_i}q_i^{t_2}, \quad t_{\text{se}} = t_2 - t, \quad t_{\text{ss}} = t - t_1 $$

$$ \Delta t = t_2 - t_1 \quad (7) $$

$$ {}^{L_i}q_i^t = \frac{t_{\text{ss}}}{\Delta t}\,{}^{L_i}q_i^{t_1} + \frac{t_{\text{se}}}{\Delta t}\,{}^{L_i}q_i^{t_2} \quad (8) $$

$$ {}^{L_i}P_i^t = \frac{t_{\text{ss}}}{\Delta t}\,{}^{L_i}P_i^{t_1} + \frac{t_{\text{se}}}{\Delta t}\,{}^{L_i}P_i^{t_2}. \quad (9) $$

The presence of NLOS invalidates some of the range information. We believe that sufficiently constrained LiDAR odometry information can satisfy the requirement of short time pose estimation. When there is no degradation [37] in the constraints obtained from the LiDAR point cloud at moments t_1 and t_2 on robot i during the matching solution and the number of constraints δ satisfies $\delta > \delta_C$, the observation can be identified as a keyframe.

Assuming a sliding window of size M, based on the structure of the factor graph, we consider that the first-order Markovian prior information through the sliding window filter acts only on the data of the first keyframe in the sliding window, i.e., the observed data at moment m. We define the function for constructing the factor graph optimization to consist of ${}_{\text{odom}}E_i^{tp}, i \in \{R_i\}$ from the LiDAR odometry, ${}_{\text{UWB}}E_{im,jn}^{tp}, i, j \in \{R_i, R_j\}, m := \{1{:}N_{\text{master}}\}, n := \{1{:}N_{\text{node}}\}$ from the UWB ranging and ${}_{\text{prior}}E_i^{tp}$ from the marginalized retention of historical observations. We define the function for constructing the factor graph optimization to consist of ${}_{\text{odom}}E_i^{tp}, i \in \{R_i\}$ from the LiDAR odometry, ${}_{\text{UWB}}E_{im,jn}^{tp}, i, j \in \{R_i, R_j\}, m := \{1{:}N_{\text{master}}\}, n := \{1{:}N_{\text{node}}\}$ from the UWB ranging and ${}_{\text{prior}}E_i^{tp}$ from the marginalized retention of historical observations. The optimization function and its expressions are shown in the following equations:

$$ {}_{\text{prior}}E_i^{tp} = \left\| Z_{R_i,P_i}^{tp}{}^{-1} \times \begin{bmatrix} {}^{W}R_i^{tp} & {}^{W}P_i^{tp} \\ 0 & 1 \end{bmatrix} \right\|_{\sum_{\text{prior}}^{-1}} \quad (10) $$

$$ {}_{\text{odom}}E_i^{tp} = \begin{cases} \left\| \log\left(\left({}^{L_i}R_i^{tp-1^T}\,{}^{L_i}R_i^{tp} \right)^{-1} \widehat{{}^{W}R_i^{tp-1}}^{T}\,\widehat{{}^{W}R_i^{tp}} \right) \right\|_{\sum_{\text{odom},R}^{-1}} \\ \left\| -{}^{L_i}R_i^{tp-1^T}\,{}^{L_i}P_i^{tp-1} + {}^{L_i}R_i^{tp-1^T}\,{}^{L_i}P_i^{tp} \right. \\ \left. \quad -\widehat{{}^{W}R_i^{tp-1}}^{T}\,\widehat{{}^{W}P_i^{tp}} + \widehat{{}^{W}R_i^{tp-1}}^{T}\,\widehat{{}^{W}P_i^{tp-1}} \right\|_{\sum_{\text{odom},P}^{-1}} \end{cases} \quad (11) $$

$$ {}_{\text{UWB}}E_{im,jn}^{tp} = \sum_{m,n=1}^{N_{\text{master}},N_{\text{node}}} \rho\left(\left\| {}_{\text{UWB}}Z_{im,jn}^{tp} - \left\| \widehat{{}^{W}R_i^{tp}}\,{}^{B_0}P_{im} + \widehat{{}^{W}P_i^{tp}} - \widehat{{}^{W}R_j^{tp}}\,{}^{B_j}P_{jn} - \widehat{{}^{W}P_j^{tp}} \right\|_{\sigma_r^{-1}} \right) \right. \quad (12) $$

$$ \widehat{X_{m:t}} = \text{argmin}\, f(X_{m:t}, Z_{1:t}) $$

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$$= \underset{X_{m:t}}{\operatorname{argmin}} \sum_{i=0:N-1,tp=m} {}_{\text{prior}}E_i^{tp} + \sum_{i=0:N-1,tp=m:t} {}_{\text{odom}}E_i^{tp} + \sum_{i=1:N-1,tp=m:t} {}_{\text{UWB}}E_{im,jn}^{tp}. \quad (13)$$

C. Relative Initialization and NLOS Detection

Fig. 3 illustrates the initialization process of co-localization, which is shown in [38]: the relative pose estimation between robots is less influenced by the initial distance, while the relative position influence is more influenced by the initial distance. We illustrate this by means of a simulation. At the uncertainty of UWB ranging of 0.05 m, i.e., $\delta_r = 0.05$ m, the interval of UWB tag located at robot master is 1 m and the height is 0.5 m, i.e.,

$${}^{B_0}P_{im}(1{:}N_{\text{master}}) = \begin{bmatrix} 0.5 & 0.5 & -0.5 & -0.5 \\ 0.5 & -0.5 & 0.5 & -0.5 \\ 0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}$$

the interval of UWB tag located in robot node is 0.5 m, and the height is 0.5, 1.5, and 0.5 m, respectively, that is,

$${}^{B_i}P_{im}(1{:}N_{\text{node}}) = \begin{bmatrix} 0.5 & 0 & -0.5 \\ 0 & 0 & 0 \\ 0.5 & 1.5 & 0.5 \end{bmatrix}.$$

Since the relative positional measurements between the systems mentioned in this article are obtained based on UWB measurements, we first numerically simulate to obtain the effect of the distance between different robot nodes on the degree of position and pose. We explored the uncertainty of the relative position and attitude measurements of the robots 0–10 m apart based on UWB in three dimensions, and the results are shown in Fig. 4 .

Next, we initialize by adding LiDAR odometry observations, and two factors need to be explored: the effect of adding LiDAR odometry and odometry degradation on the relative position initialization. We assume that the LiDAR odometry may be degraded by underobservation in the vertical direction, so we add $\sum_{\text{odom1}} = 0.1 * I, \sum_{\text{odom2}}$ for 0.2 m in the vertical direction, respectively, to the simulation. For the sake of generality, we simulate the trajectory of robot node i in a circular motion of radius 10 m around robot master.

The simulation results in Fig. 5 verify the effect of Z-axis degradation on the relative position, especially on the relative position estimation, and demonstrate the importance of fusing two valid measurement information. We consider that the initial value of relative pose can get using a set of effective LiDAR odometry measurement and UWB ranging measurement. The effectivity of LiDAR odometry measurement can be determined by the $\delta_{\text{init}-c}$ number of constraints constructed by feature of point cloud from each robot node or robot master. Another principle is that the eigenvalues $\lambda_i, i := 1{:}6$ of the Fisher information matrix [37] of LiDAR odometry features used for the solution must be greater than λ_{threld}. As shown in

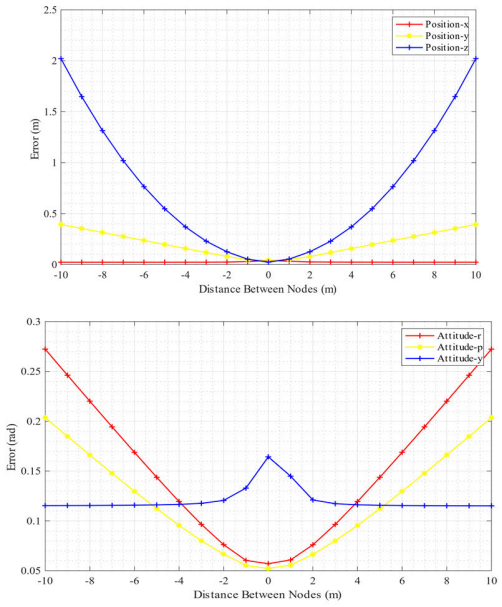


Fig. 4. Effect of distance and relative position uncertainty and relative attitude uncertainty. (a) Relative position uncertainty. (b) Relative attitude uncertainty. The three curves are the errors in the three vertical directions of *XYZ* in turn.

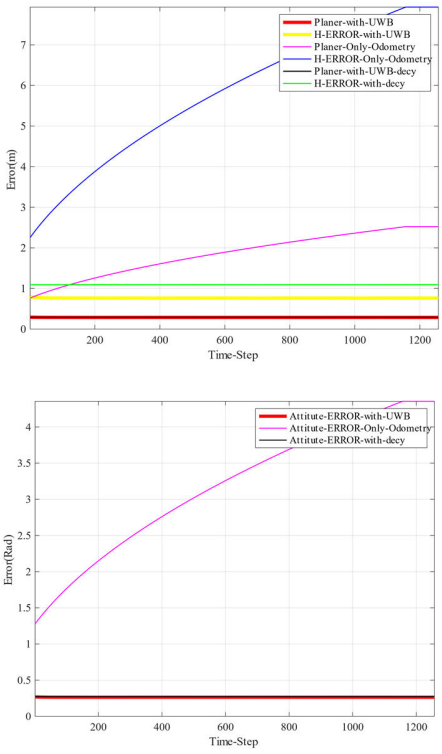


Fig. 5. Effect of LIDAR odometry Z-axis information degradation on relative pose estimation. (a) Position drift profiles, six curves are horizontal and vertical drift without degradation, horizontal and vertical drift without ranging constraint, and horizontal and vertical drift with Z-axis degradation. (b) Attitude drift, three curves are relative attitude drift curves without degradation, relative attitude drift curves without ranging constraint, and relative attitude drift curves with Z-axis degradation.

Fig. 6, an initialization equation for solving the relative poses is constructed in the following equation:

$$\left\{ {}^{L_0}P_{L_1}^{t_m}, \ldots, {}^{L_0}\widehat{P_{L_1}^{t}}, {}^{L_0}P_{L_2}^{t_m}, \ldots, {}^{L_i}P_{L_2}^{t} \right\}$$

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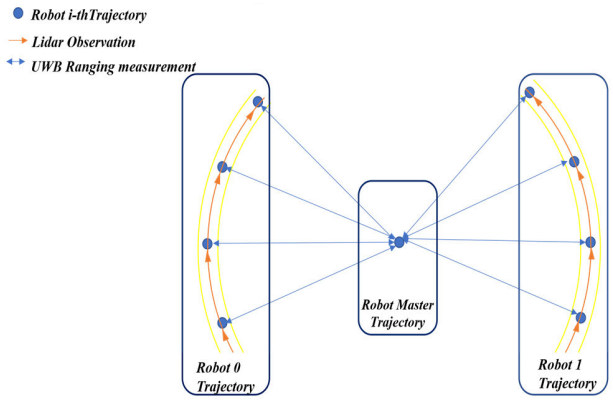


Fig. 6. Schematic of the initialization factor map. The factors used for optimization consist of range and odometry constraints and plane constraints. The variables used for optimization are the relative poses between the initialization keyframes.

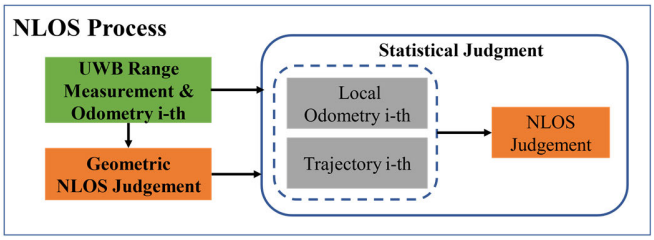


Fig. 7. NLOS processing flow: the UWB range information is first judged by the UWB using the range triangle structure, and then, it is finally judged whether it is NLOS according to the relative positional results of the initialization process.

$$= \underset{\left\{{}^{L_0}P^{tm}_{L_1},\ldots,{}^{L_0}P^{t}_{L_1},{}^{L_0}P^{tm}_{L_2},\ldots,{}^{L_i}P^{t}_{L_2}\right\}}{\operatorname{argmin}} \sum_{i=0:N-1,tp=m:t} {}_{\text{odom}}E^{tp}_{i} + \sum_{i=1:N-1,tp=m:t} {}_{\text{UWB}}E^{tp}_{im,jn}. \tag{14}$$

NLOS is unavoidable in the process of UWB measurements. In this article, we use a combined geometric and statistical approach to determine whether UWB distance measurements are NLOS. Fig. 7 illustrates the judgment process of NLOS in this article, where the geometric judgment method, as shown in Fig. 8, makes full use of the triangular geometric features of the baseline construction. However, when two robot nodes are far away from each other, the geometry-based judgment method can hardly meet the discrimination requirements.

We assume that any two UWB tag of robot i, i.e., ${}^{B_i}P_{im}$ and ${}^{B_i}P_{in}$ have a baseline of $b_{im,in}$, the ranging information of UWB tag k from robot j is defined as $r_{jk,im}$ and $r_{jk,in}$, then the ranging will have the following geometry constraints:

$$\begin{cases} \left\| \|r_{jk,im}\| + \|r_{jk,in}\| \right\| \geq \left\| b_{im,in} \right\| \\ \left\| \|r_{jk,im}\| - \|r_{jk,in}\| \right\| \leq \left\| b_{im,in} \right\|. \end{cases} \tag{15}$$

However, geometry-based judgment methods are difficult to satisfy the discrimination requirements when two robot nodes are far away. Therefore, we need to use a statistical algorithm [39] for NLOS judgments. The latest odometry measurement can be used to predict the robot pose as the initial value for the NLOS detection

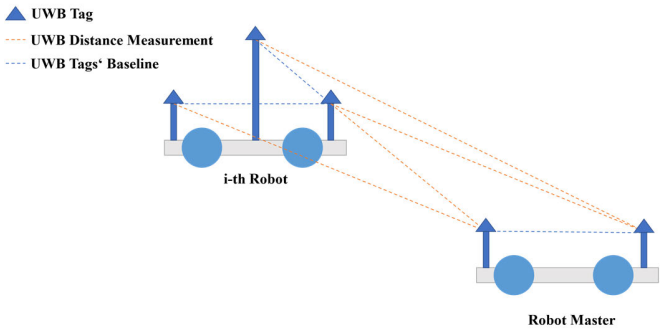


Fig. 8. Distance measuring triangle diagram. The principle according to which the sum of two sides in a triangle is greater than the third side and the sum of two sides is less than the third side. UWB tags' baseline refers to the distance measurement between different UWB tags on the same robot.

$$\begin{cases} {}^{W}R^{t}_{i} = \widehat{{}^{W}R^{t-1}_{i}} \times {}^{L_i}R^{t-1^{T}}_{i} \times {}^{L_i}R^{t}_{i} \\ {}^{W}P^{t}_{i} = \widehat{{}^{W}R^{t-1}_{i}} \times {}^{L_i}R^{t-1^{T}}_{i} \times {}^{L_i}P^{t}_{i} - \widehat{{}^{W}R^{t-1}_{i}} \times {}^{L_i}R^{t-1^{T}}_{i} \\ \qquad \times {}^{L_i}P^{t-1}_{i} + \widehat{{}^{W}P^{t-1}_{i}}. \end{cases} \tag{16}$$

LiDAR odometry has fine relative pose resolution performance, and we believe that this initial value can be applied to the NLOS discrimination. Therefore, based on the discrimination criteria of [39], the second NLOS judgment is performed as follows:

$$\left\| r^{t}_{im,jn} - \left\| {}^{W}R^{t}_{i} \times {}^{B_i}P_{im} + {}^{W}P^{t}_{i} - {}^{W}R^{t}_{j} \times {}^{B_j}P_{jn} - {}^{W}P^{t}_{j} \right\| \right\| < 2\sigma_r. \tag{17}$$

D. Relative Initialization and NLOS Detection

When the co-localization and NLOS judgments are finished, the key frame data can be used for co-building maps. The information used to pass into the central robot in Fig. 1 includes local map information. The point clouds composing the local map information are mainly the noise removed and distortion corrected point clouds corresponding to the LiDAR odometry ${}^{Li}R^{t}_{i}, {}^{Li}P^{t}_{i}$ output moment $t, {}^{Li}M^{t}_{i}$. The distortion correction and filtering methods are like the methods in [40]. Fig. 9 illustrates the directed graph model composed of collaborative mapping. At this stage, the size of the window sliding is relatively large because relatively accurate positional estimation results can be obtained. As shown in Fig. 9, the accuracy of the constructed map is dependent on the degree of co-location accuracy, and the constructed map is optimally aligned to the global coordinate system $\{W\}$ based on the second map of the poses.

The distance constraint still produces an uncertainty of 0.05 m, i.e., $\delta_{r2} = 0.05$ m, and according to the improvement of the original odometry by the co-located pose solution, the odometry $\sum_{O2} = \sum_{O}/2$. The observations added to this optimization process become based on the global coordinate system $\{W\}$. The optimization method for directed graph constraints in this section is based on ISAM2 [41]. GTSAM [42] can simplify and apply the method of [41]. Then, the pdf for

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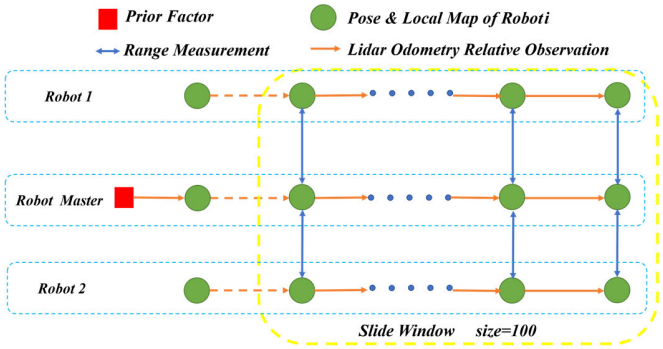


Fig. 9. Based on the sliding window collaborative mapping factor graph, the estimated values optimized by iSAM2 using collaborative localization are used in the article as the initial values of the collaborative mapping Bayesian filter.

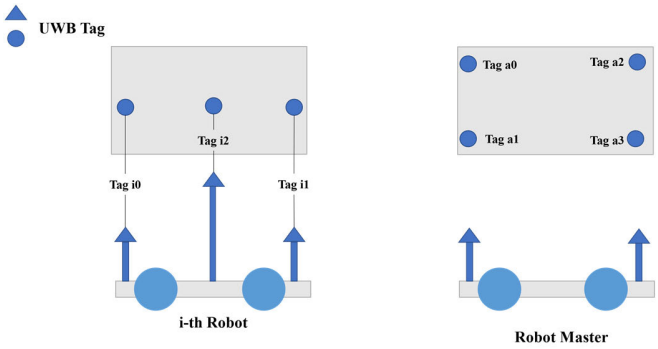


Fig. 10. Illustration of the UWB tags arrangement. The master node has four UWB tags, and it is essential to investigate the effect of the tags from the UWBs on the nodes on the relative pose estimation.

this graph optimization is

$$P(X_{t1:t1+99}|Z_{1:t1+99}, X_1) = \eta P\left(X_{t1:t1+99}|Z_{t1:t1+99}, X_{t1}^0\right) \times P\left(X_{t1}^0|X_1^0, Z_{1:t1-1}\right). \quad (18)$$

After reconstructing the collaborative mapping observation, the optimization function used to build the map is constructed as follows:

$${}_{\text{prior}}E_i^{tp} = \left\| Z_{R_i,P_i}^{tp\ -1} \times \begin{bmatrix} {}^W R_0^{tp} & {}^W P_0^{tp} \\ 0 & 1 \end{bmatrix} \right\|_{\sum_{\text{prior}}^{-1}} \quad (19)$$

$${}_{\text{odom}}E_i^{tp} = \begin{cases} \left\| \log\left(\left({}^W R_i^{tp-1\,T}\, {}^W R_i^{tp} \right)^{-1} \widehat{{}^W R_i^{tp-1}}^{\,T} \widehat{{}^W R_i^{tp}} \right) \right\|_{\sum_{\text{odom2},R}^{-1}} \\ \left\| -{}^W R_i^{tp-1\,T}\, {}^W P_i^{tp-1} + {}^W R_i^{tp-1\,T}\, {}^W P_i^{tp} \right. \\ \left. \quad -\widehat{{}^W R_i^{tp-1}}^{\,T} \widehat{{}^W P_i^{tp}} + \widehat{{}^W R_i^{tp-1}}^{\,T} \widehat{{}^W P_i^{tp-1}} \right\|_{\sum_{\text{odom2},P}^{-1}} \end{cases} \quad (20)$$

$${}_{\text{UWB}}E_{im,jn}^{tp} = \sum_{m,n=1}^{N_{\text{master}},N_{\text{node}}} \rho\left(\left\| {}_{\text{UWB}}Z_{im,jn}^{tp} - \left\| \widehat{{}^W R_i^{tp}}\, {}^{B_0}P_{im} + \widehat{{}^W P_i^{tp}} - \widehat{{}^W R_j^{tp}}\, {}^{B_j}P_{jn} - \widehat{{}^W P_j^{tp}} \right\| \right\|_{\sigma_{r2}^{-1}} \right) \quad (21)$$

$$\begin{aligned} \widehat{X_{t1:t1+99}} &= \operatorname{argmin} f(X_{t1:t1+99}, Z_{1:t1+99}) \\ &= \sum_{i=0,tp=t1} {}_{\text{prior}}E_i^{tp} + \sum_{i=0:N-1,tp=t1:t1+99} {}_{\text{odom}}E_i^{tp} \\ &\quad + \sum_{i=1:N-1,tp=t1:t1+99} {}_{\text{UWB}}E_{im,jn}^{tp}. \end{aligned} \quad (22)$$

E. Tag Count and Fisher Matrix Analysis

The relative positional estimation errors arising from UWB tag distance and degradation are explored in Section III-C. This section uses simulations to illustrate the effect of tag number and high difference of robot node on relative positional estimation. In [43] and [44], the Cramér–Rao lower bound (CRLB) of the relative pose estimation and Fisher information matrix (EIM) for 2-D robots and distributed vehicles is

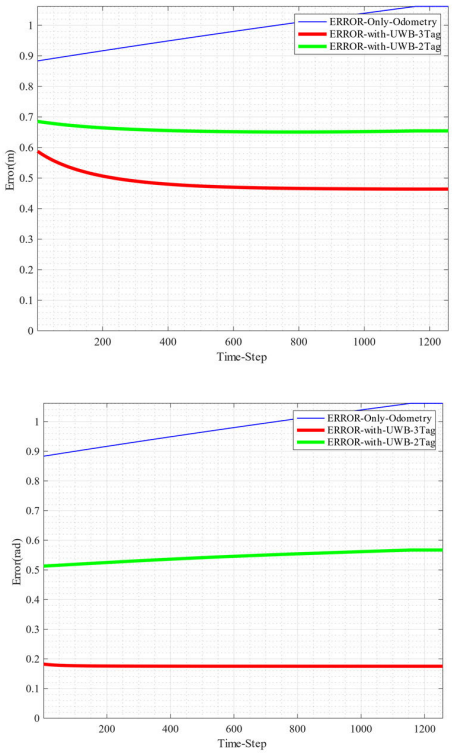


Fig. 11. Effect of the number of UWB tag on the relative position estimation. (a) Relative position influenced by the number of UWB tag. (b) Relative attitude influenced by the number of UWB tag. The blue curve shows the error when only the LiDAR odometry is used, green shows the error in the presence of two UWB tags, and red shows the error in the presence of three UWB tags.

analyzed by theoretical analysis plus numerical simulations. The CRLB of the variable X is denoted by $I(X)$

$$I(X) = E\left\{ \left(X - \widehat{X}\right)\left(X - \widehat{X}\right)^T \right\}. \quad (23)$$

For the observation function $Z(X)$, $n_Z \sim N(\widehat{Z}, \sum_Z)$, the FIM is defined as $(\partial Z/\partial X)^T \sum_Z{}^{-1} (\partial Z/\partial X)$. The experimental conditions assumed are consistent with the simulation test conditions of Section III-C; here, $\sum_{\text{odom}} = 0.01\ \text{m} * I$. The deployment of UWB in the master and slave nodes is shown in Fig. 10. We first verified the effect of the number of UWB tag on the accuracy of relative position and relative attitude estimation. We tested the effect of the presence of four UWB

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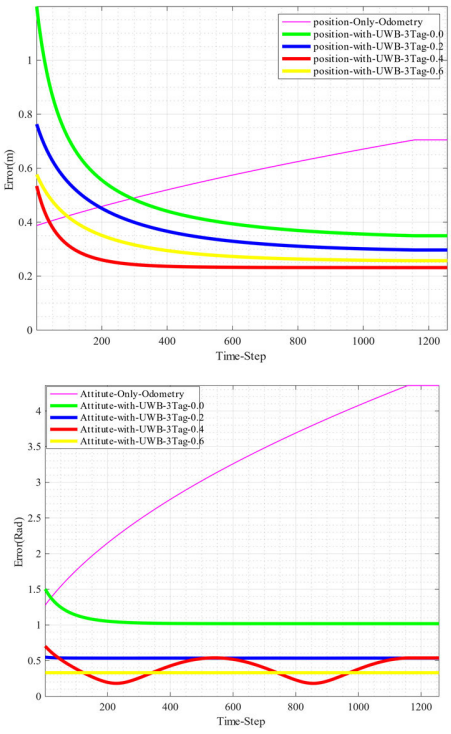


Fig. 12. Effect of height difference of robot node tag on relative positioning and relative attitude estimation. (a) Relative position estimation. (b) Relative attitude estimation. The magenta curve shows the error when only the LiDAR odometry is used, green shows the error when the height difference between the third UWB Tag and the other two tags is 0.0 m, blue shows the error when the height difference between the third UWB tag and the other two tags is 0.2 m, red shows the error when the height difference between the third UWB tag and the other two tags is 0.4 m, and yellow shows the error when the height difference between the third UWB tag and the other two tags is 0.4 m. UWB tag and the other two tags at a height difference of 0.6 m.

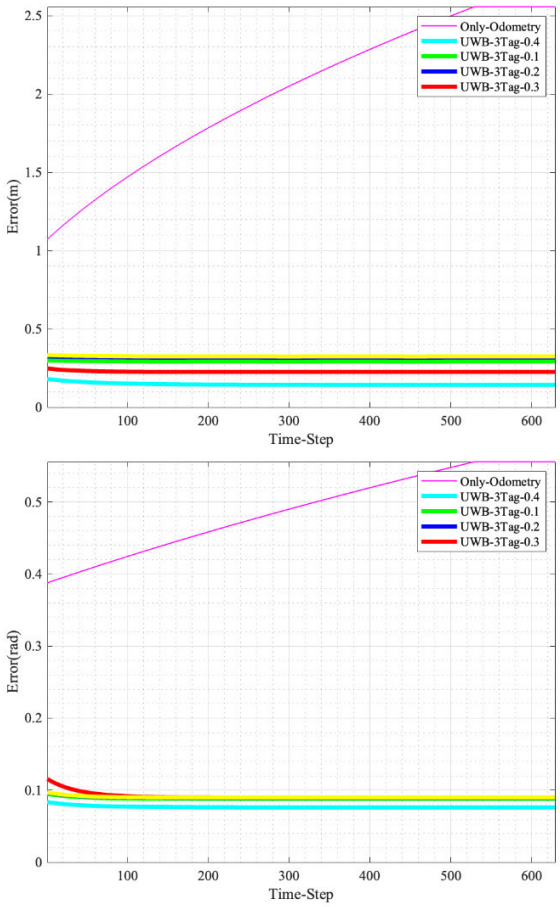


Fig. 13. Z-axis drift suppression capability. (a) Relative position estimation. (b) relative attitude estimation. The magenta curve shows the error when only the LiDAR odometry is used. The sky blue, green, blue, and red curves show the error when the height difference between the third UWB Tag and the other two UWB Tags is 0.4, 0.1, 0.2, and 0.3 m, in that order.

tags in robot master and 0, 2, and 3 UWB tags in robot node on the relative localization accuracy, respectively. We put the a priori factor $\sum_{\text{prior}} = 0.5 * I$. The test results are shown in Fig. 11.

The algorithm mentioned in this article is applied to ground swarm, but the ground swarm is vulnerable to Z-axis drift in the direction of vertical ground. Therefore, we investigated the effect of the relative positioning accuracy in the presence of three UWB tags, so that the height difference between the tags in the direction of vertical ground has an impact on the relative positioning accuracy, and we let the third UWB tag of robot node in turn at the height difference of 0, 0.2, 0.4, and 0.6 m. The results are shown in Fig. 12.

From the results of the numerical simulation, we chose an arrangement with a height difference of 0.4 m. We explore the degree of suppression of Z-axis drift by this arrangement. We set the degree of Z-axis drift to 0.4, 0.1, 0.2, and 0.3 m, respectively for numerical simulation, and the simulation results are shown in Fig. 13. In addition, we explored the extent to which the window size affects the relative position estimation. We verified the estimation errors for window sizes of 0, 10, 50, and 100, respectively, and the numerical results are shown in Fig. 14.

Based on the numerical results, we believe that the height difference and the actual degree of observation noise and degradation of odometry have a significant degree of influence on the relative pose estimation. Therefore, we chose an altitude difference setting of 0.4 m for the next practical test procedure and selected a LiDAR odometry with relatively stable performance.

IV. Experiments

This part verifies the feasibility of our proposed algorithm by an actual UGV team. As shown in Fig. 15, we built the testbed of the multirobot. The robot team mainly consists of three robots. The team communicates with each other through a local area network. Each robotics team member contains a Velodyne VLP16 LiDAR, which has an output frequency of 10 Hz. The output frequency of the UWB ranging information is 50 Hz. The processor of each robot is NAVIDA Jetson AGX Xavier with ARMv8.2 CPU and 2.26 GHz, and the computer carries Ubuntu 18.04. The positional truth data are obtained by RTK, which has a position estimation accuracy of 1 cm and an attitude accuracy of 1°. The LiDAR odometry on each robot is F-LOAM [45], and the output frequency of this odometry can reach up to 30 Hz. Practical tests by Wang et al. [45] show that the uncertainty of this odometry is: $\sum_{\text{odom},R} = 0.0048° * I$ and $\sum_{\text{odom},P} = 0.002 \text{ m} * I$. Section IV-A tested the effect of the system's

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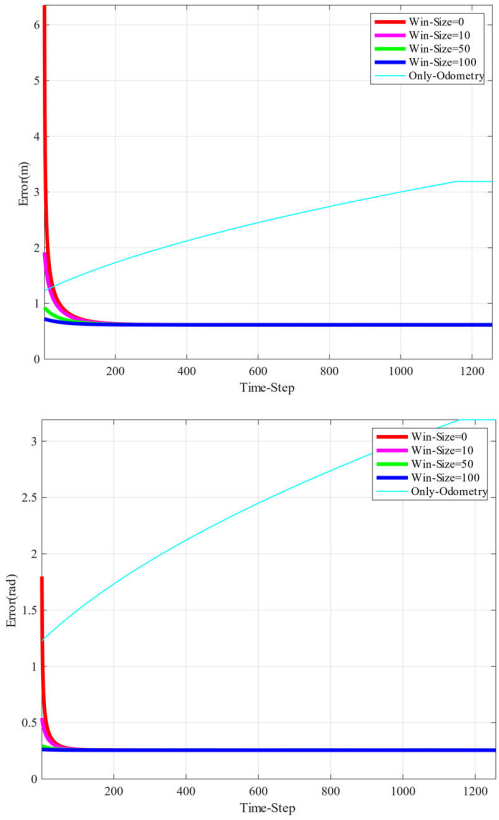


Fig. 14. Effect of window size on the degree of relative positional estimation. (a) Relative position estimation. (b) Relative attitude estimation. The sky blue curve shows the error when using only the LiDAR odometry, while the red, magenta, green, and blue curves show the error for window sizes of 0, 10, 50, and 100, in that order.

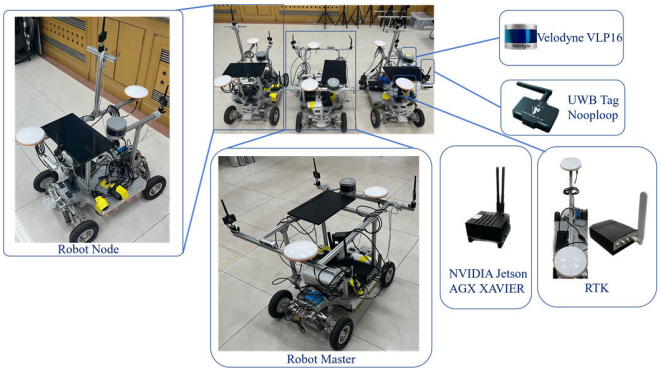


Fig. 15. Experimental equipment.

NLOS processing capability and the number of UWBs on co-location. Section IV-B tests the co-localization capability of the system, and this part also verifies the effectiveness of the fusion of range information and LIDAR odometry information. Section IV-C tests the system's collaborative mapping capability.

As shown in Fig. 16, the coordinate system $\{L_i\}$ of the LIDAR overlaps with the body coordinate system $\{B_i\}$ of the robot. Four UWB tags exist on the robot master. Three UWB tags exist on the robot node. The reason for using four tags for the robot master is the flip ambiguity of the relative pose as appeared in [19]. The position of the UWB tag on the robot member in the robot body coordinate system is defined

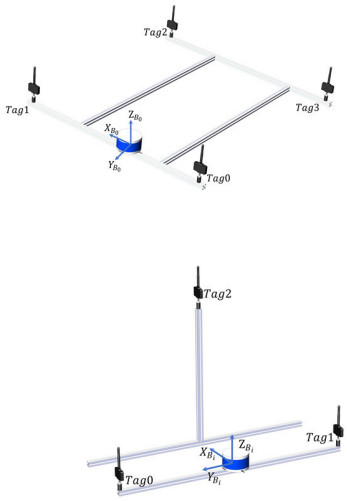


Fig. 16. Body frame of robot master and robot node. (a) Robot master' body frame. (b) Robot node' body frame.

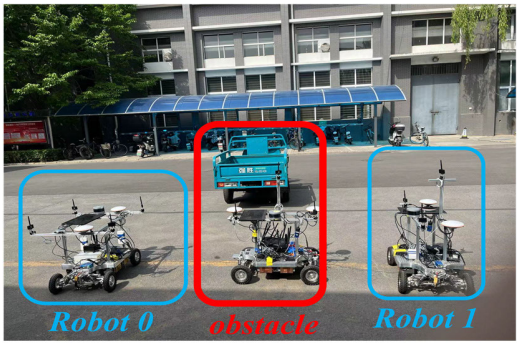


Fig. 17. NLOS presence scenario, where robot 2 is treated as an obstacle in this test, and another normal cart is treated as an obstacle with both robot 2.

as UWB extrinsic, ${}^{B_i}P_{im}$, $i \in \{0{:}2\}$, $m \in \{1{:}\text{Node}_i\}$, where the UWB tag extrinsic on robot master, ${}^{B_0}P_{im}$,

$$m \in \{1{:}4\} = \begin{bmatrix} -0.24 & 0.44 & 0.44 & -0.24 \\ 0.0 & 0.0 & -0.68 & -0.68 \\ -0.03 & -0.03 & -0.03 & -0.03 \end{bmatrix} \text{m}$$

and the extrinsic parameter of robot node is ${}^{B_i}P_{im}$, $i \in \{1{:}2\}$,

$$m \in \{1{:}3\} = \begin{bmatrix} 0.0 & 0.0 & 0.20 \\ -0.34 & 0.34 & 0.01 \\ -0.03 & -0.03 & 0.36 \end{bmatrix} \text{m}.$$

A. NLOS Condition and Range Information Test

Our initialization method and the initialization are based on the running trajectory. We explore the effect of three trajectories on the initialization. They are circular motion, isotropic parallel motion, and reverse parallel motion. Based on the three running trajectories, we explored the effect of different distances on UWB tags' number.

As shown in Fig. 17, we have two obstacles in this scene, and they are robot 2 at rest and the tricycle at rest. Fig. 18 shows the comparison between just going through the baseline and using the NLOS method proposed in this article. First, in the environment where obstacles are present, the range

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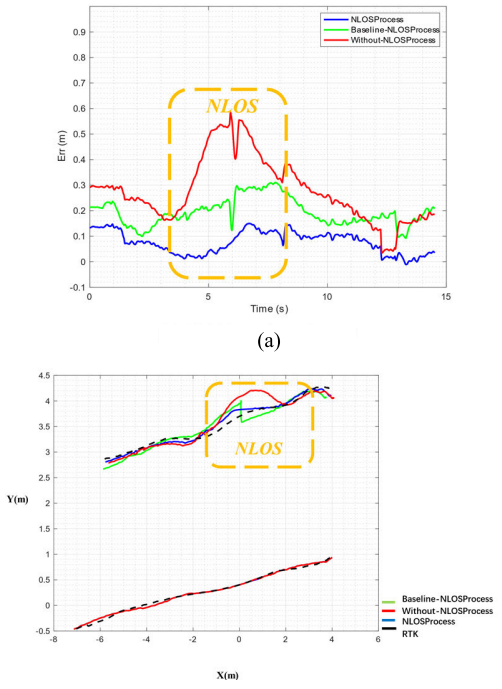


Fig. 18. NLOS processing. (a) Results of co-localization error in NLOS scenario at the running moment. The red curve is the result without NLOS processing, the green curve is the result using only the baseline processing method, and the blue curve is the result of NLOS processing proposed in this article. (b) Two-vehicle trajectory map in NLOS scenario.

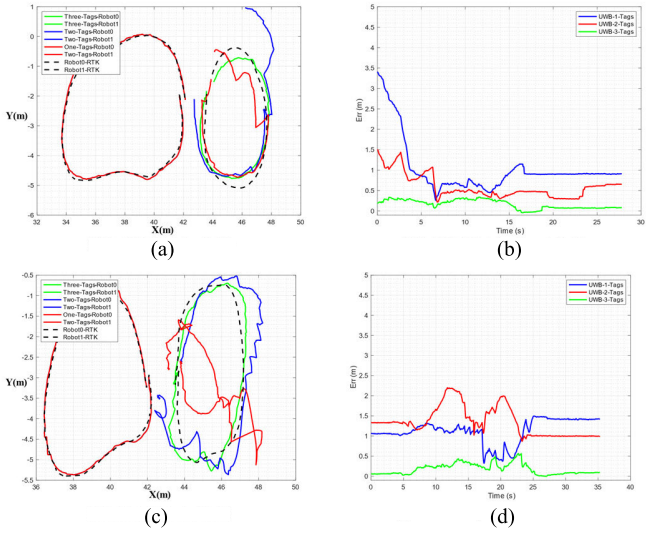


Fig. 19. Results of two-vehicle trajectory estimation. (a) and (c) shows the trajectory estimation results for the number of tag from 1 to 3, respectively. (b) and (d) shows the trajectory estimation error of robot 1. The black dashed line is the true value of the RTK trajectory estimate, and the blue curve and the red and green curves are the estimates when only one UWB tag is present.

measurement produces outliers. This section mainly shows the ability of the cooperative localization system to handle NLOS in the environment where NLOS is generated by the system. The results of this test also verify that the two-step NLOS-based determination in this article brings higher robustness and accuracy to the system than the geometric

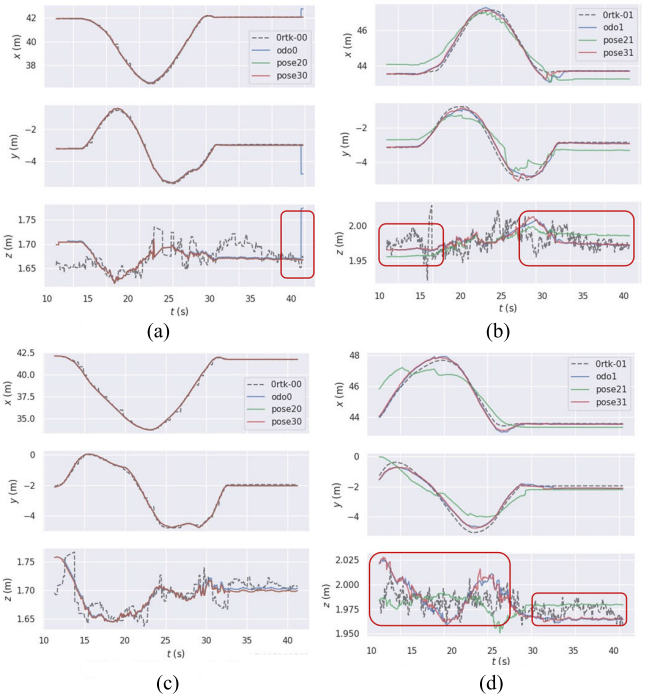


Fig. 20. Degree of drift suppression of the *x*-*y*-*z* axes by the number of UWB tags. The black curve is the trajectory estimation from RTK, the blue curve is the estimation result from LiDAR odometry, the green curve is the co-estimation result when robot node has two UWB tags, and the red curve is the co-estimation result when three UWB tags are present. The drift comparison of several results is shown in the red boxed area. (a) Robot 0 Trajectory Test01. (b) Robot 1 Trajectory Test01. (c) Robot 0 Trajectory Test02. (d) Robot 1 Trajectory Test02.

TABLE I
UWB TAG COUNT-DUAL VEHICLE TRAJECTORY ESTIMATION ERROR

Test 01				Test 02			
2 Tags		3 Tags		2 Tags		3 Tags	
Robot 0	Robot 1	Robot 0	Robot 1	Robot 0	Robot 1	Robot 0	Robot 1
0.09m	0.49m	0.09m	0.17m	0.11m	1.08m	0.11m	0.15m

baseline-based approach and the case of no NLOS processing. In the following, we tested the effect of the number of UWB distance measurements, i.e., the number of range tags, on the co-location accuracy. We performed the test with a two-vehicle system. Fig. 19 verifies the effect of the number of range information on the accuracy of the trajectory solution by trajectory solving. Fig. 19 shows the results of the two tests. We tested the scenarios with the number of tags from 1 to 3, respectively. Table I shows the co-localization accuracy for both tests. Estimation accuracy of co-localization. The trajectory estimation results of robot 0 are basically the same in the process of each test, which indicates that the selection of LiDAR odometry is more reasonable in this test. Therefore, the richness of UWB ranging information has a major impact on the results of co-localization.

In addition, we verified the effect of our method on Z-axis drift with this test. We tested the degree of trajectory drift in each coordinate axis direction. As shown in Fig. 20, we compared the estimated results of the LiDAR odometry with the drift degree for tag numbers 2 and 3, respectively. This test demonstrates the reasonableness of our UWB arrangement.

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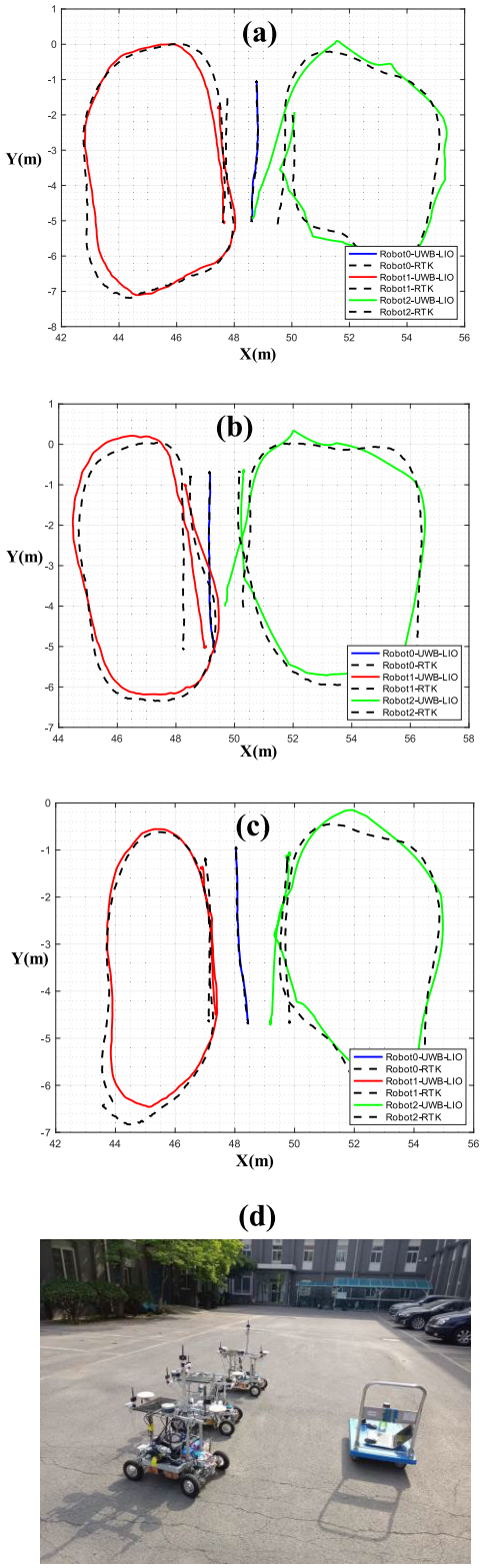


Fig. 21. Co-location results of the three-robot team. (a)–(c) are the results of the UWB-LO cooperative trajectory estimation tests for the three tests, respectively. The last figure shows the test scenario machine equipment. The black dashed line is the real value measured by RTK, the blue curve is the estimated trajectory of robot master, the red curve is the estimated trajectory of Robot1, and the green curve is the estimated trajectory of Robot2. (d) Test scene.

The arrangement of UWB tag with height difference helps to suppress the drift in the Z-axis.

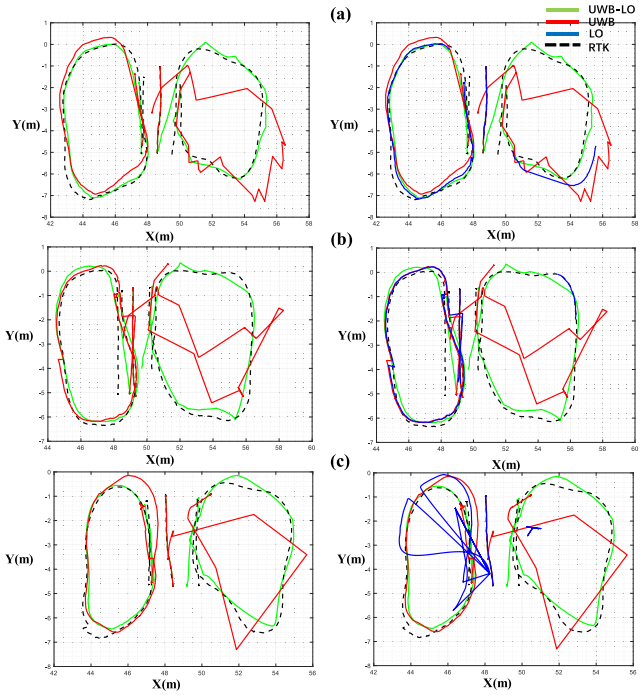


Fig. 22. Comparison results of UWB-Only, LiDAR-odometry-only, and UWB-LO-fusion. The black dashed line is the true value of the RTK measurement, and the green curve is the synoptic estimated trajectory of the UWB and LiDAR odometry fusion. The red curve is the co-estimated trajectory when only UWB range information is used. The blue curve is the estimated trajectory of the LiDAR odometry. (a) Test01. (b) Test02. (c) Test03.

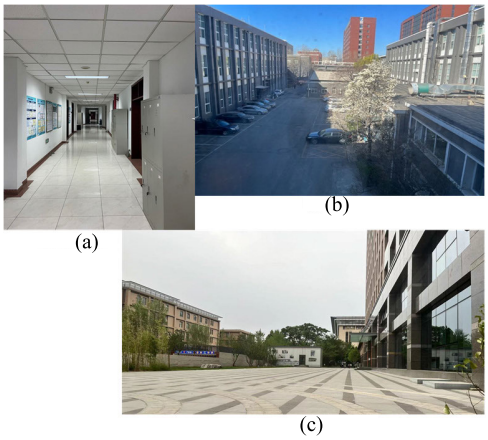


Fig. 23. Experiment scene. (a) Scene 01. (b) Scene 02. (c) Scene 03.

TABLE II
CO-LOCATION RESULTS OF THE THREE-ROBOT TEAM ESTIMATION ERROR

Test 01			Test 02			Test 03		
Robot 0	Robot 1	Robot 2	Robot 0	Robot 1	Robot 2	Robot 0	Robot 1	Robot 2
0.09 m / 0.67°	0.21m/ 3.44°	0.35m/ 4.01°	0.08 m / 0.69°	0.32 m / 4.77°	0.62 m / 6.81°	0.04 m / 0.53°	0.72 m / 4.84°	0.85m /5.66°

B. Cooperative Localization

In this section, the system mainly demonstrates its own collaborative localization capability. The system is a team of three robots that perform cooperative trajectory estimation in a 14 m $*$ 8 m space. There are four UWB tags on robot

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TABLE III
CO-LOCATION ACCURACY EVALUATION

Data	Robot 0				Robot 1				Robot 2			
	LO	UWB	EKF-Fusion	UWB-LO	LO	UWB	EKF-Fusion	UWB-LO	LO	UWB	EKF-Fusion	UWB-LO
Test 01	0.12m/0.74°	0.092m/0.82°	0.094m/1.42°	**0.090m/0.67°**	0.23m/6.49°	0.39m/9.04°	0.36m/7.91°	**0.21m/3.44°**	/	2.45m/3.44°	3.95m/5.41°	**0.35m/4.01°**
Test 02	**0.071m/0.60°**	0.08m/0.79°	0.08m/0.66°	0.08m/0.69°	0.33m/6.31°	0.34m/5.93°	0.322m/5.93°	**0.32m/4.77°**	/	7.02m//	8.44m//	**0.62m/6.81°**
Test 03	0.13m/0.77°	0.04m/0.71°	0.047m/1.33°	**0.04m/0.53°**	/	2.34m/9.93°	/	**0.72m/4.84°**	/	9.25m//	/	**0.85m/5.66°**

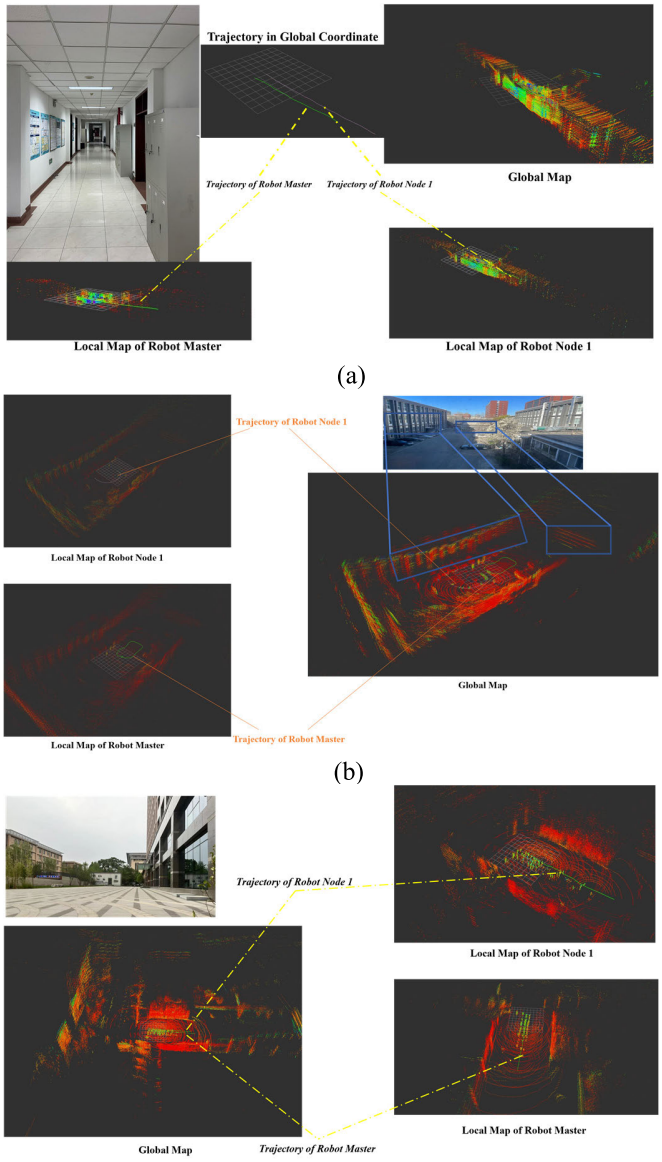


Fig. 24. Cooperative reconstruction of two ground robot based on the proposed framework. The figure shows the collaborative construction of three scenarios, which shows the dual vehicle trajectory of robot master and node 1 (trajectory of robot master and trajectory of robot node 1) in the three scenarios and the sub map (local map of robot master and local map of robot node 1) based on a single vehicle at a single moment and their global map (global map) constructed collaboratively in real time. (a) Test scene 01. (b) Test scene 02. (c) Test scene 03.

master, which are located at the same horizontal plane with a spacing of 0.68 m. There are three UWB tags on robot node, where the height of the third UWB tag is higher than the other two by 0.4 m. As shown in Fig. 21, we show the results of cooperative trajectory estimation for three tests. The effect of this estimation is shown in Tables II and III.

In this experiment, we found that differences in trajectory estimation accuracy emerged in the same scenarios. We tried to investigate the reasons for such differences. We compared the scenarios using UWB alone and LIDAR odometry alone with the scenario where the two are fused, and the results of the comparison are shown in Fig. 22. The odometry used for co-estimation in test 1/2 showed relatively reliable results during the robot team estimation, and the results in test 3 showed a significant drift, making the change information less reliable. In addition, the LiDAR odometry was utilized to a greater extent in robot 0/1 than in robot 2 during test 1/2, so robot 0/1 was estimated to be more accurate than robot 2. From the results of test 3, although the LiDAR odometry produced a large degree of drift, the addition of the LiDAR odometry measurements helped the overall co-trajectory estimation in terms of the degree of increase in the amount of information in the information matrix.

C. Cooperative Mapping

We tested the collaborative mapping capability of the system through real-time mapping of three scenes. As shown in Fig. 23, the three scenes are a narrow indoor corridor, a parking lot surrounded by buildings, and a small square in front of a building. Due to the bandwidth limitation of the wireless router we used and the need to implement transmission, the local submaps built on robot node are sparse LiDAR point clouds. The results of the three scenes built are shown in Fig. 24. The construction of point cloud maps based on positional estimation does not require restrictions based on the common observation area, which can avoid the serious consequences of submap point cloud matching errors and can reduce the cost of map.

V. CONCLUSION

In this article, a mapping localization system based on a graph optimization model for the fusion of UWB ranging and LiDAR odometry information is proposed. The fusion of UWB range information with odometry pose estimation information and mapping information can accomplish collaborative mapping by the robot team. The presence of ranging information can suppress the degradation of the odometry information to some extent, especially in some directions where the observations are degraded. The cooperative positioning system based on the initialization of relative poses enables the positioning system to converge faster for real-time positioning and mapping. The number of UWB tags and the arrangement on the robot have a large impact on the

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cooperative positional estimation, especially the relative positional estimation. We believe that the arrangement of UWB ranging tag with certain height difference can produce effective suppression of odometry drift. In addition, in future research, we would like to explore the effectiveness of synergy between ground and air clusters and synergy of observation information based on 3-D free-moving air clusters.

REFERENCES

[1] Z. Li, C. Jiang, X. Gu, Y. Xu, F. Zhou, and J. Cui, "Collaborative positioning for swarms: A brief survey of vision, LiDAR and wireless sensors based methods," *Defence Technol.*, May 2023.

[2] Y. Xianjia, L. Qingqing, J. P. Queralta, J. Heikkonen, and T. Westerlund, "Applications of UWB networks and positioning to autonomous robots and industrial systems," in *Proc. 10th Medit. Conf. Embedded Comput. (MECO)*, Jun. 2021, pp. 1–6.

[3] A. Wasik, R. Ventura, J. N. Pereira, P. U. Lima, and A. Martinoli, "LiDAR-based relative position estimation and tracking for multi-robot systems," in *Proc. 2nd Iberian Robot. Conf. (Robot)*. Springer, 2016, pp. 3–16.

[4] P. Schmuck and M. Chli, "CCM-SLAM: Robust and efficient centralized collaborative monocular simultaneous localization and mapping for robotic teams," *J. Field Robot.*, vol. 36, no. 4, pp. 763–781, 2019.

[5] Z. Xun et al., "CREPES: Cooperative relative pose estimation system," 2023, *arXiv:2302.01036*.

[6] M. Karrer, M. Agarwal, M. Kamel, R. Siegwart, and M. Chli, "Collaborative 6DoF relative pose estimation for two UAVs with overlapping fields of view," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 6688–6693.

[7] R. Jung, C. Brommer, and S. Weiss, "Decentralized collaborative state estimation for aided inertial navigation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 4673–4679.

[8] M. Saska, J. Vakula, and L. Preucil, "Swarms of micro aerial vehicles stabilized under a visual relative localization," in *Proc. IEEE Int. Conf. Robot. Automat. (ICRA)*, Hong Kong, 2014, pp. 3570–3575, doi: 10.1109/ICRA.2014.6907374.

[9] T. Nguyen, K. Mohta, C. J. Taylor, and V. Kumar, "Vision-based multi-MAV localization with anonymous relative measurements using coupled probabilistic data association filter," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 3349–3355.

[10] H. Xu, L. Wang, Y. Zhang, K. Qiu, and S. Shen, "Decentralized visual-inertial-UWB fusion for relative state estimation of aerial swarm," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2020, pp. 8776–8782.

[11] F. Zhu et al., "Swarm-LIO: Decentralized swarm LiDAR-inertial odometry," 2022, *arXiv:2209.06628*.

[12] T. Hoang Nguyen and L. Xie, "Relative transformation estimation based on fusion of odometry and UWB ranging data," 2022, *arXiv:2202.00279*.

[13] T. Ziegler, M. Karrer, P. Schmuck, and M. Chli, "Distributed formation estimation via pairwise distance measurements," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3017–3024, Apr. 2021.

[14] C. C. Cossette, M. Shalaby, D. Saussié, J. R. Forbes, and J. Le Ny, "Relative position estimation between two UWB devices with IMUs," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4313–4320, Jul. 2021.

[15] X. Yu, P. T. Morrn, S. Salimpour, J. P. Queralta, and T. Westerlund, "Loosely coupled odometry, UWB ranging, and cooperative spatial detection for relative Monte-Carlo multi-robot localization," 2023, *arXiv:2304.06264*.

[16] Z. Cao et al., "Relative localization of mobile robots with multiple ultra-WideBand ranging measurements," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 5857–5863.

[17] K. Guo, Z. Qiu, W. Meng, L. Xie, and R. Teo, "Ultra-wideband based cooperative relative localization algorithm and experiments for multiple unmanned aerial vehicles in GPS denied environments," *Int. J. Micro Air Veh.*, vol. 9, no. 3, pp. 169–186, 2017.

[18] B. Jones, K. Thompson, D. Pierce, S. Martin, and D. Bevly, "Ground-vehicle relative position estimation with UWB ranges and a vehicle dynamics model," *IFAC-PapersOnLine*, vol. 55, no. 37, pp. 681–687, 2022.

[19] M. Shalaby, C. C. Cossette, J. R. Forbes, and J. Le Ny, "Relative position estimation in multi-agent systems using attitude-coupled range measurements," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4955–4961, Jul. 2021.

[20] A. Fishberg and J. P. How, "Multi-agent relative pose estimation with UWB and constrained communications," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2022, pp. 778–785.

[21] M. Karrer, P. Schmuck, and M. Chli, "CVI-SLAM—Collaborative visual-inertial SLAM," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 2762–2769, Oct. 2018, doi: 10.1109/LRA.2018.2837226.

[22] P. Schmuck, T. Ziegler, M. Karrer, J. Perraudin, and M. Chli, "COVINS: Visual-inertial SLAM for centralized collaboration," in *Proc. IEEE Int. Symp. Mixed Augmented Reality Adjunct (ISMAR-Adjunct)*, Bari, Italy, Oct. 2021, pp. 171–176, doi: 10.1109/ISMAR-Adjunct54149.2021.00043.

[23] P. Zhu, Y. Yang, W. Ren, and G. Huang, "Cooperative visual-inertial odometry," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Xi'an, China, May 2021, pp. 13135–13141, doi: 10.1109/ICRA48506.2021.9561674.

[24] T. Zhang, L. Zhang, Y. Chen, and Y. Zhou, "CVIDS: A collaborative localization and dense mapping framework for multi-agent based visual-inertial SLAM," *IEEE Trans. Image Process.*, vol. 31, pp. 6562–6576, 2022, doi: 10.1109/TIP.2022.3213189.

[25] L. Bernreiter, S. Khattak, L. Ott, R. Siegwart, M. Hutter, and C. Cadena, "Collaborative robot mapping using spectral graph analysis," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, Philadelphia, PA, USA, May 2022, pp. 3662–3668, doi: 10.1109/ICRA46639.2022.9812102.

[26] I. Deutsch, M. Liu, and R. Siegwart, "A framework for multi-robot pose graph SLAM," in *Proc. IEEE Int. Conf. Real-Time Comput. Robot. (RCAR)*, Angkor Wat, Cambodia, Jun. 2016, pp. 567–572, doi: 10.1109/RCAR.2016.7784092.

[27] R. Dube, A. Gawel, H. Sommer, J. Nieto, R. Siegwart, and C. Cadena, "An online multi-robot SLAM system for 3D LiDARs," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Vancouver, BC, Canada, Sep. 2017, pp. 1004–1011, doi: 10.1109/IROS.2017.8202268.

[28] Y. Xie et al., "RDC-SLAM: A real-time distributed cooperative SLAM system based on 3D LiDAR," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 9, pp. 14721–14730, Sep. 2022, doi: 10.1109/TITS.2021.3132375.

[29] Y. Huang, T. Shan, F. Chen, and B. Englot, "DiSCo-SLAM: Distributed scan context-enabled multi-robot LiDAR SLAM with two-stage global-local graph optimization," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 1150–1157, Apr. 2022.

[30] S. Zhong, Y. Qi, Z. Chen, J. Wu, H. Chen, and M. Liu, "DCL-SLAM: A distributed collaborative LiDAR SLAM framework for a robotic swarm," 2022, *arXiv:2210.11978*.

[31] A. Papadimitriou, S. S. Mansouri, and G. Nikolakopoulos, "Range-aided ego-centric collaborative pose estimation for multiple robots," *Exp. Syst. Appl.*, vol. 202, Sep. 2022, Art. no. 117052.

[32] R. Liu et al., "Distributed ranging SLAM for multiple robots with ultra-WideBand and odometry measurements," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Kyoto, Japan, Oct. 2022, pp. 13684–13691, doi: 10.1109/IROS47612.2022.9982028.

[33] Y. Song, M. Guan, W. P. Tay, C. L. Law, and C. Wen, "UWB/LiDAR fusion for cooperative range-only SLAM," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, Montreal, QC, Canada, May 2019, pp. 6568–6574, doi: 10.1109/ICRA.2019.8794222.

[34] F. J. Perez-Grau, F. Caballero, L. Merino, and A. Viguria, "Multi-modal mapping and localization of unmanned aerial robots based on ultra-wideband and RGB-D sensing," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Vancouver, BC, Canada, Sep. 2017, pp. 3495–3502, doi: 10.1109/IROS.2017.8206191.

[35] Y. Cao and G. Beltrame, "VIR-SLAM: Visual, inertial, and ranging SLAM for single and multi-robot systems," *Auto. Robots*, vol. 45, no. 6, pp. 905–917, Sep. 2021.

[36] H. Zhou, Z. Yao, Z. Zhang, P. Liu, and M. Lu, "An online multi-robot SLAM system based on LiDAR/UWB fusion," *IEEE Sensors J.*, vol. 22, no. 3, pp. 2530–2542, Feb. 2022, doi: 10.1109/JSEN.2021.3136929.

[37] J. Zhang, M. Kaess, and S. Singh, "On degeneracy of optimization-based state estimation problems," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2016, pp. 809–816.

[38] S. Zheng et al., "UWB-VIO fusion for accurate and robust relative localization of round robotic teams," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 11950–11957, Oct. 2022, doi: 10.1109/LRA.2022.3208354.

[39] S. Zheng, Z. Li, Y. Liu, H. Zhang, and X. Zou, "An optimization-based UWB-IMU fusion framework for UGV," *IEEE Sensors J.*, vol. 22, no. 5, pp. 4369–4377, Mar. 2022, doi: 10.1109/JSEN.2022.3144660.

[40] J. Zhang and S. Singh, "LOAM: LiDAR odometry and mapping in real-time," in *Proc. Robot., Sci. Syst.*, 2014, vol. 2, no. 9, pp. 1–9.

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[41] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. J. Leonard, and F. Dellaert, "iSAM2: Incremental smoothing and mapping using the Bayes tree," *Int. J. Robot. Res.*, vol. 31, no. 2, pp. 216–235, Feb. 2012.
[42] F. Dellaert, "Factor graphs and GTSAM: A hands-on introduction," Georgia Inst. Technol., Atlanta, GA, USA, Tech. Rep. GT-RIMCP&R-2012-002, Sep. 2012.
[43] Y. Liu, Y. Wang, J. Wang, and Y. Shen, "Distributed 3D relative localization of UAVs," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 11756–11770, Oct. 2020.
[44] X. Shen, Y. Liu, and Y. Shen, "On the spatial information coupling in relative localization networks," in *Proc. IEEE Int. Conf. Commun.*, Montreal, QC, Canada, Jun. 2021, pp. 1–6, doi: 10.1109/ICC42927.2021.9500501.
[45] H. Wang, C. Wang, C.-L. Chen, and L. Xie, "F-LOAM : Fast LiDAR odometry and mapping," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Prague, Czech Republic, Sep. 2021, pp. 4390–4396, doi: 10.1109/IROS51168.2021.9636655.



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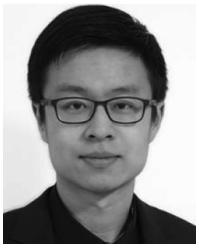
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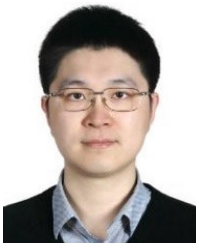
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